

Extreme events and subtle ecological effects: lessons from a long-term sugar maple–American beech comparison

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Abstract. Increasing extreme events that are related to global change are expected to affect the dynamics of forest ecosystems. If disruptive stressors (e.g., insects, drought) affect tree vigor without causing mortality, the ecological effects may be subtle, making subsequent ecosystem dynamics more difficult to predict than in the case of disturbances causing death. Based on the literature and our personal observations, we expected that such a subtle change could have occurred in the dynamics between sugar maple and American beech. We implemented a targeted paired-sampling design (1) to verify whether a change occurred (gradual or abrupt, recovered or not) in the growth dynamics between the two species over a 57-yr period, (2) to identify the likely causes of this change, and (3) to investigate whether such changes could trigger other long-time ecological consequences. We found that sugar maple growth was negatively affected by an extreme event (or a few events) between 1986 and 1989, while American beech was not affected. Twenty years after the 1986–1989 abrupt growth decrease, sugar maple (1) had a slower growth than American beech, although it was previously similar, (2) did not respond to monthly climatic variations as it did prior to the abrupt growth decrease, and (3) had lower resilience when faced with a new stress event. Overall, our study, besides showing that extreme events with subtle effects may change the dynamics of an ecosystem, also illustrates that these events may accelerate ecosystem misadaptation to climate. Fine-scale targeted monitoring is essential to complement broad-scale monitoring to detect such misadaptations in a global change context.

Key words: American beech growth; compound ecological effect; drought; ecosystem dynamics; sugar maple growth; thaw–freeze event.

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INTRODUCTION

Global change, changes in natural disturbance regimes (Kurz et al. 2008), increases in extreme climatic events (Bell et al. 2004), and the arrival of new pathogens (Ricciardi 2007) are expected to affect forest ecosystem dynamics in a variety of ways. When events kill trees suddenly (e.g., fire or wind), subsequent forest dynamics are fairly predictable and can be relatively easy to model (Foster 1988, Kruger and Reich 1997). In

contrast, if the disruptive agents (e.g., insects, drought) affect tree vigor without killing them, their ecological effects may be expressed more subtly. Subsequent ecosystem dynamics are then more difficult to predict because many interacting plant physiological processes are involved (Rosenzweig et al. 2001, McDowell et al. 2011). Incidentally, researchers have recently become interested in pervasive small-scale mortality events, such as increased drought-induced mortality (Choat et al. 2012,

Allen et al. 2015) when these events are scaled over large areas and long time frames, they can affect a large number of trees (Bourque et al. 2005, Peng et al. 2011). Moreover, these subtle effects may last for long periods of time, depending on tree vigor at the moment when a stress occurs (Camarero et al. 2015, Mamet et al. 2015) or if multiple stressors are implicated (Gutschick and BassiriRad 2003, Denny et al. 2009). In a context where we wish to predict the effects of climate change and favor ecosystem adaptation to it (Millar et al. 2007, West et al. 2009), it is essential to understand how species and individuals recover following such non-destructive, disruptive events.

The sugar maple (*Acer saccharum* Marsh.)–American beech (*Fagus grandifolia* Ehrh.) ecosystem is particularly useful for posing such a question. It is generally agreed that these two long-lived and late-successional species co-exist; that is, neither species seems to supplant the other (Poulson and Platt 1996). Following non-destructive disruptive events, we should thus be able to detect whether there is a change in the dynamics between the two species and possible recovery. Further, there is an extensive literature on growth decline in sugar maple that has been associated with acid rain (Juice et al. 2006), drought events (Allen et al. 1992, Gavin et al. 2008), thaw–freeze events (Payette et al. 1996) or insects (Bauce and Allen 1991, Hartmann and Messier 2008), which suggests that dynamics between the two species could have changed in the recent decades to favor beech. In addition, several studies (Beaudet et al. 1999, Duchesne et al. 2005, Messier et al. 2011) have already identified a change in the dynamics between the two species at the regeneration stage.

Thus, in this article, we wanted (1) to verify whether a change had occurred (gradual or abrupt, recovered or not) in the growth and subsequent species dynamics between two co-existing late-successional species, viz. sugar maple and American beech; (2) to identify the likely causes of this change; and (3) to investigate whether such changes could trigger other long-term ecological consequences. We thus established a unique sampling design of 150 pairs of sugar maple and American beech across all size classes to compare the long-term growth of the two species.

MATERIALS AND METHODS

Study area

This study was conducted in the Papineau-Labelle Wildlife Reserve (46°13'48" W, 75°09'55" N) of Quebec, between Lakes Montjoie and Du Sourd, and about 100 km northeast of Canada's capital, Ottawa. The area is located in the eastern portion of Lac du Poisson Blanc landscape unit (Robitaille and Saucier 1998) of the western sugar maple-yellow birch (*Betula alleghaniensis* Britton) bioclimatic region (Saucier et al. 1998). The landscape contains numerous hills with elevations <450 m a.s.l. and averaging 300 m in height. Mean annual temperature is 3.7°C, mean annual precipitation is roughly 1100 mm (including 250 mm as snow), and the number of degree-days above 0°C is 2716 (Environment Canada 2017). Surficial geology for the study area is characterized by thin to moderately thin glacial till composed of metamorphic rocks, such as gneiss, topped by sandy Dystric Brunisols (Soil Landscapes of Canada Working Group (2010). A study conducted in similar stands in the same region (Nolet et al. 2008) showed that mean pH generally is 5.1 and that mean Ca concentration is 410 mg/kg. The forest canopy is dominated by sugar maple in association with yellow birch, American beech, American basswood (*Tilia americana* L.), ironwood (*Ostrya virginiana* (Mill.) K.Koch), eastern hemlock (*Tsuga canadensis* (L.) Carr.), and balsam fir (*Abies balsamea* (L.) Mill.).

Tree sampling

Ten (10) sugar maple dominated stands, with no tree harvesting in the last 20 yr, were selected in the autumn of 2007. In each of these stands, we sampled 16 sugar maple and 16 American beech, which were divided into four major classes of dbh 10–18 cm; 18.1–26 cm; 26.1–34 cm; 34.1–42 cm; because some pairs could not be found, our sampling led to a total of 150 (instead of 160) pairs of the two species. To minimize the effects of the growing environment (in terms of light and soil conditions) when comparing both species growth, individuals were sampled as part of a pair, one sugar maple and one American beech. The stands we sampled were not infected by the beech bark disease; this is noteworthy to mention as this disease complex—between an insect and a

pathogen—may affect American beech growth (Kasson and Livingston 2012, Nolet et al. 2015). We considered individuals as a pair if they met the following criteria:

1. To have <2 cm dbh difference (in 2007);
2. To be part of the same dominance class (suppressed, intermediate, co-dominant, or dominant)
3. To exhibit similar competition from surrounding trees in 2007, measured using a factor 2 (metric) prism sweep;
4. To be far enough from one another to avoid direct competition for light;
5. To be close enough to each other (e.g., 25 m) to reasonably assume they grew under similar edaphic conditions.

In addition, a minimum 100 m distance was respected between each pair of trees. For each tree, a core was sampled to the pith of the tree (unless trees were rotten) using an increment borer.

Core samples were mounted and then sanded using increasingly fine sandpaper grits. The ring widths from 1950 to 2007 were measured to the nearest 0.01 mm using a 40× magnification scope and a sliding measurement stage (Velmex, Bloomfield, New York, USA), which was coupled to a digital meter. Quality control of cross-dated series was checked using both graphical and statistical methods provided by the *dplR* package (Bunn 2010) in R (version 3.1.0; R Core Team 2015), which integrates many of the functionalities of commonly used dendrochronological software, such as COFECHA (Holmes 1983).

Data analysis

In order to check for changes in sugar maple and beech growth over time (1950–2007), we first averaged the radial growth of each species for each year. Second, we calculated the difference in radial growth between sugar maple and American beech for each pair of trees and for each year. We then calculated the mean difference (and 95% confidence interval) between both species for each year. We visually inspected the radial growth patterns of both species for reduced tree-ring patterns that could be associated with potential stressors: insects, droughts, and thaw–freeze events. As the forest tent caterpillar (FTC) is known to feed on sugar maple, we used the

data collected for the study region (Cooke and Lorenzetti 2006) to identify possible FTC outbreaks. Drought and thaw–freeze events were identified using daily data from the closest weather station (Mont-Laurier; Environment Canada 2017). We considered a drought event to occur when the total amount of precipitation over a 21-continuous day period (from May to August) was 5 mm or less. As a comparison, the mean daily precipitation for the region in summer is 3 mm. Six drought events were identified between 1950 and 2007 (Appendix S1: Fig. S1). A winter thaw–freeze event compounds a warm (usually >0°C) period followed by an abrupt drop in temperature (Pockman and Sperry 1997, Bourque et al. 2005). To identify the most important thaw–freeze events over the 1950–2007 period, we plotted each individual day (x_i) from January and February of each year according to (1) the thaw length, that is, the number of consecutive warm days (reaching at least 0°C) and (2) the temperature drop, that is, the difference between the maximum temperature of x_i and the minimum temperature of x_{i+1} . From this analysis, we decided to retain the two most obvious thaw–freeze events (Appendix S1: Fig. S2). For both drought and thaw–freeze, threshold was determined to identify extreme events during the period of interest. These thresholds do not imply that the events necessary had physiological effects; in the same way, it is possible that other events not meeting our definition of drought or thaw–freeze induced physiological effects on (some) trees. However in both cases, it is likely that our thresholds induced physiological effects (Bourque et al. 2005, Galvez et al. 2011).

To evaluate the relationship between climate and species growth, we first calculated a detrended growth index for both species. This detrending process was accomplished in two steps (Mériani and Lebourgeois 2011), again using the *dplR* package (Bunn 2010). First, we fitted a 20-yr cubic smoothing spline with a 50% frequency response on each raw measurement series (individuals). For each species, residuals from the fitting procedure then were averaged by year using a bi-weighted robust mean to obtain a growth index (Appendix S1: Fig. S3). It is worth mentioning that this growth index was obtained using trees of all stands pooled together as we were looking for a global signal in all trees and

not a stand specific signal and thus analyses were not conducted at the stand level. Furthermore, trees were located at a minimum distance of 100 m to avoid local effects. To verify whether the relationship between climatic variable changed after the 1986–1989 abrupt growth decrease, we first performed simple correlation analyses between monthly temperature, precipitation and water balance (obtained through BioSIM; Régnière and Saint-Amant 2008), and growth indices for both species and for two periods: 1950–1985 and 1990–2005 (i.e., before the event of 2006–2007 for this specific analysis). Climate-growth relationships were performed using both current and previous climatic variables. For further regression analyses, we kept only variables that appeared to have an influence on species growth during 1950–1985. To ensure that the best relationships between climate and growth for 1950–1985 were not artifacts of a bigger dataset (36 yr compared to 16 for the 1990–2005 period), we performed a resampling analysis, in which 1000 samples of 16 random years (between 1950 and 1985) were generated. Multiple regression analyses were performed on each of the 1000 samples, and the confidence intervals of the adjusted R^2 were compared to the R^2 that was obtained using all years. This analysis of the stationary, or non-stationary, effect of climatic variables on growth is inspired from usual stationary analyses, but it permitted us (1) to directly test whether there is a change in the climate-growth relationship after a specific event, (2) to analyze periods that differ in their lengths, and (3) to incorporate multiple regression into the analysis.

Finally, to verify the presence of a link between the 1986–1989 drop and the one observed in 2006–2007, we calculated the relative drop in maple growth compared to the growth in the years preceding the drop for each individual:

$$\begin{aligned} \text{Relative growth drop in 1986-1989 (RDG}_1\text{)} \\ = (G_{1986-1989} - G_{1982-1985}) / (G_{1982-1985}) \end{aligned} \quad (1)$$

and

$$\begin{aligned} \text{Relative growth drop in 2006-2007 (RDG}_2\text{)} \\ = (G_{2006-2007} - G_{2004-2005}) / (G_{2004-2005}) \end{aligned} \quad (2)$$

where G is the mean radial growth for the identified period. We then performed a simple regression between RDG_1 and RDG_2 . Equations (1) and

(2), which lead to a relative value between 0 and 1, have been specifically used to remove the difference in growth response that may be due to dbh or age. All statistical analyses were performed in R (version 3.1.0; R Core Team 2015).

RESULTS

Radial growth of the two late-successional species, sugar maple and beech, was very similar from 1950 to 1985 (Fig. 1A), except for a few years around 1955. From 1975 to 1982, there is a drop in growth for both species, possibly due to three successive droughts from 1975 to 1978. Afterward, there is an initial sharp decline in the growth of maple in 1986, both in absolute terms and relative to beech. A second drop is observed in 1988 followed by a weak recovery in 1990. Between 1990 and 2005, the growth of sugar maple never recovered to a rate comparable with that of beech (Fig. 1A, B). These growth fluctuations are observed in all sampled diameter size classes (Fig. 1C) and can also be observed no matter whether growth is expressed in radial growth (Fig. 1) or basal area growth (Appendix S1: Fig. S4). The severe abrupt growth decrease for bigger stems cannot be attributed to a size-effect or a natural stand dynamics effect, as it is observed across size classes (Fig. 1C). Furthermore, an additional dataset (Appendix S1: Fig. S5) shows that 40-cm dbh sugar maples from the same sites had a 2.3 mm/yr radial growth around 1975, whereas current radial growth of similar-sized SM is only 1 mm/yr. The abrupt nature of sugar maple growth decline suggests a cause that is related to an event (or series of events) rather than to self-thinning or tree age. Furthermore, the fact that the abrupt growth decrease can be observed in the whole study region suggests that the event is weather related or due to a regional disturbance rather than to a localized disturbance. After this period of recovery, a further drop in maple growth was observed in 2006 and 2007.

Numerous stress events occurred during the study period, including outbreaks of forest tent caterpillar (*Malacosoma disstria*), droughts and thaw-freeze events (Fig. 1A). The 1986–1989 sugar maple abrupt growth decrease appeared to be first associated with a major thaw-freeze event that was observed in January 1986, and which

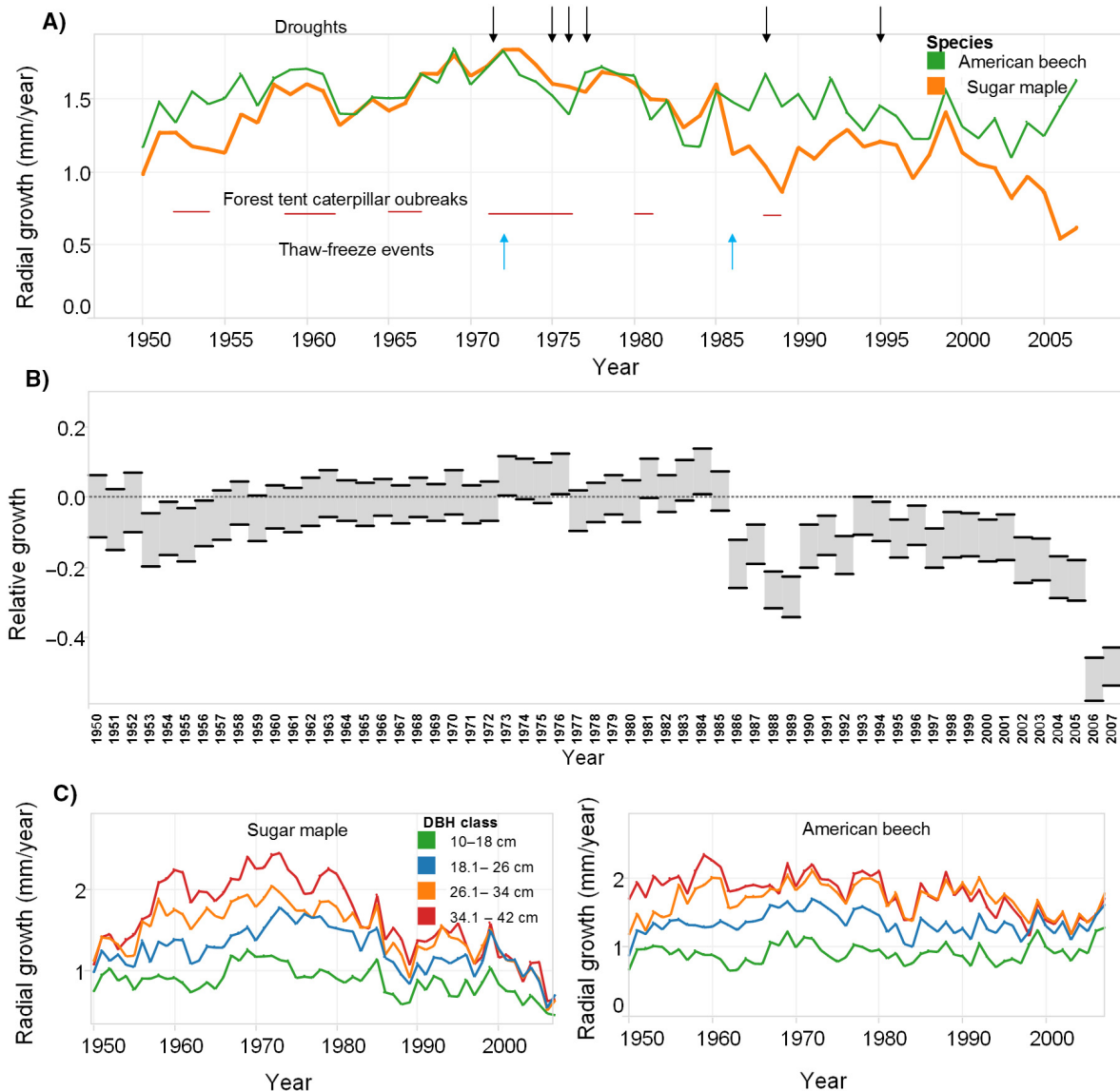


Fig. 1. Radial growth comparison between sugar maple and American beech over a 57-yr period; (A) absolute values, (B) relative values, a negative value signifies lower growth for sugar maple; the gray boxes indicate 95% confidence intervals. (C) Mean sugar maple and American beech radial growth by dbh class. Black and blue arrows (panel A), respectively, indicate drought and thaw–freeze events, as indicated by daily meteorological data (see *Materials and methods* and Appendix S1). Red lines (panel A) show documented forest tent caterpillar outbreaks (see *Materials and methods*).

was later exacerbated by a drought event in 1988 and possible forest tent caterpillar defoliation.

Prior to 1986, the radial growth of sugar maple and beech followed one another closely (Fig. 1A). After 1989, their radial growth patterns differed. This result, when coupled with the inability of sugar maple growth to recover to the same level

as beech after 1989 (Fig. 1B), clearly reveals a change in the dynamics between the two species. Regardless of the stage considered, beech was less susceptible to the stress that occurred during the 1986–1989 period than was sugar maple.

We found a surprisingly weak association between the two species growth indices after

1989 (Fig. 1 and Appendix S1: Fig. S3) and investigated it. Prior to 1986 (1950–1986), climatic variables and sugar maple growth were clearly related, as total rainfall in June and July and June temperature explained nearly 38% (adjusted R^2) of variation in the growth index of sugar maple. For the period 1990–2005, adjusted R^2 drops to 0 (Fig. 2). Resampling analysis (see *Methodology*) ensured that this result was not due simply to a difference in the number of years between the two periods; the confidence interval for the adjusted R^2 after resampling was between 0.39 and 0.41. For beech, the relationship with climatic variables was already weaker before 1986 and remained low after 1989. None of the multiple regression analyses explained further growth variation. This supports decoupling of sugar maple growth and climatic variables after 1989, as the capacity of sugar maple stems to take advantage of most favorable climatic conditions likely decreased.

The latest drop in sugar maple growth, which was observed in 2006–2007 (for which no precise cause could be identified), also highlights the

potentially long lasting effects of the 1986–1989 stressors. When the relationship between the sugar maple abrupt growth decrease in 2006–2007 is compared with that documented in 1986–1989 (Fig. 3), we observe that the trees most affected in 1986–1989 are those most vulnerable in 2006–2007. This is especially true for larger stems (Fig. 3), indicating that growth recovery is more difficult for bigger (and older) stems. As a result of this lack of recovery, the difference in growth rates between sugar maple and beech tends to shift to the benefit of beech.

DISCUSSION

The sugar maple abrupt growth decrease between 1986 and 1989 observed in this study has been reported across its range (Payette et al. 1996, Duchesne et al. 2002, Gavin et al. 2008, Hartmann and Messier 2008, Bishop et al. 2015); yet, no consensus exists regarding its causes. This study is the first one to show that the growth of American beech, the main companion species of sugar maple, was not affected during this period.

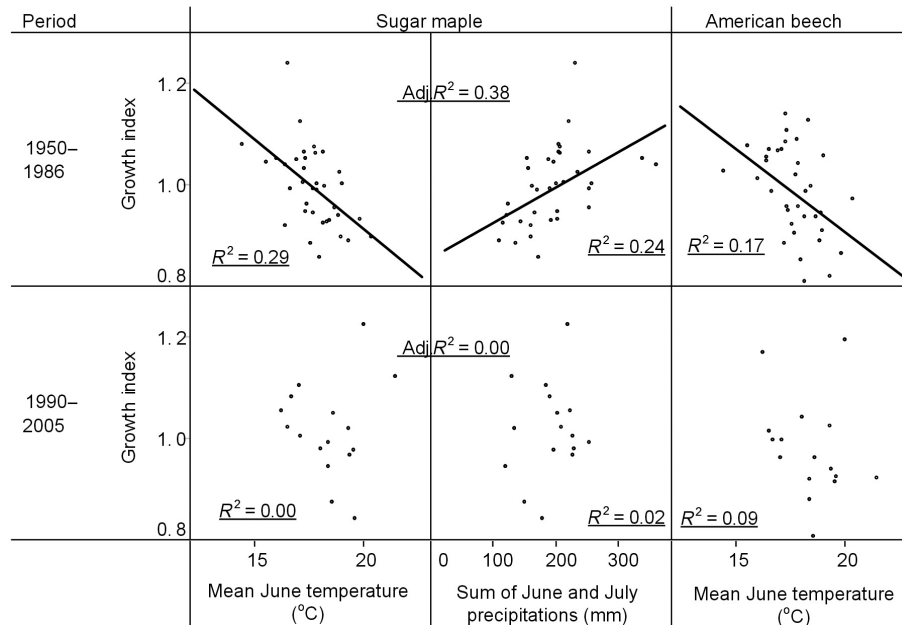


Fig. 2. Sugar maple and American beech growth indices as a function of mean June temperature and June–July precipitation. R^2 -values are provided for simple regressions, while adjusted R^2 are provided for multiple regressions. None of the multiple regression analyses for American beech are shown, as they did not explain any further variation in growth. Trend lines show significant ($P < 0.05$) simple linear relationships.

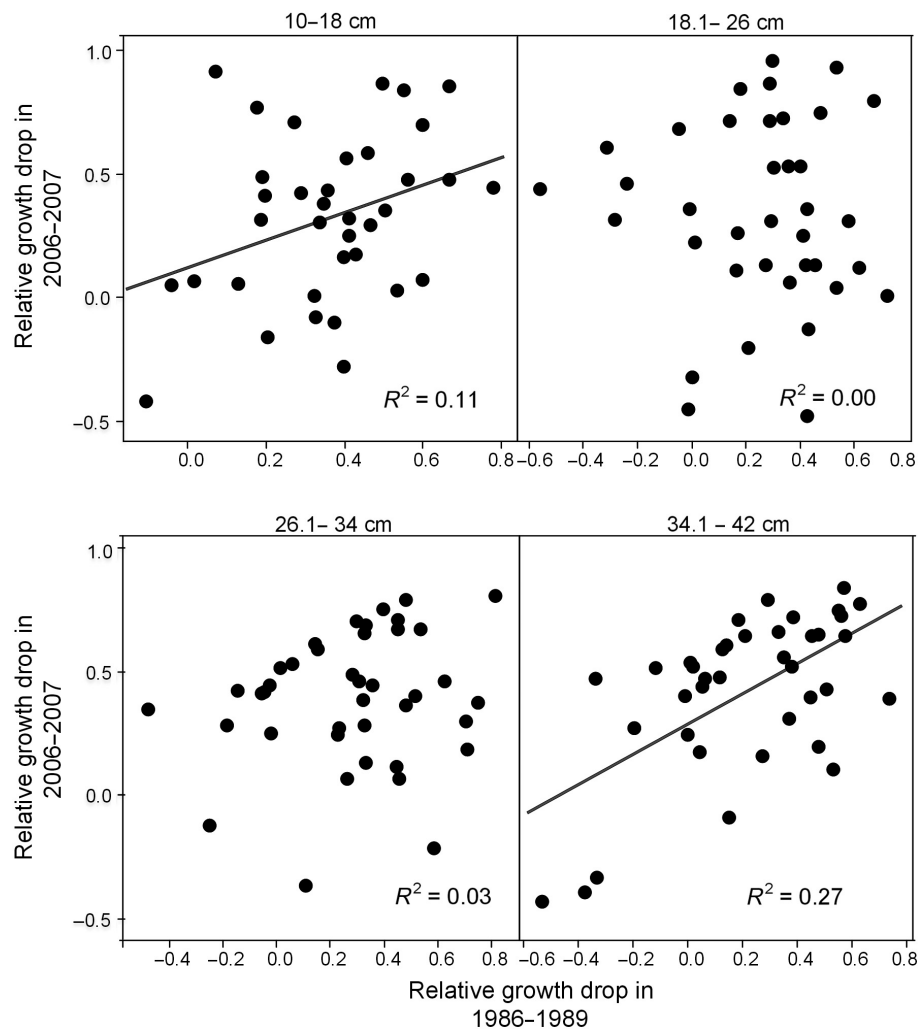


Fig. 3. Relative sugar maple growth drop in 2006–2007 as a function of the relative growth drop in 1986–1989 by dbh classes. Trend lines show significant ($P < 0.05$) simple linear relationships. See *Materials and methods* for details on relative growth radial calculations.

The reasons behind the contrasting susceptibilities of the two species during the period 1986–1989 also remain obscure, even though some hypotheses can be advanced. For example, the strong xylem refilling capacity of sugar maple that allows it to recover from winter embolism (Hacke and Sperry 2001) could be activated too rapidly in the case of an intense thaw–freeze event and lead to xylem cavitation. Trees are generally sensitive to drought following cavitation induced by thaw–freeze events (Auclair et al. 1996) as well as being more vulnerable to attack by insects. It is also possible that beech is better

adapted to water stress because of efficient stomatal control of xylem embolism (Lemoine et al. 2002), or because of a higher stem flow (Frost and Levia 2014). Another possible hypothesis is that soil acidification weakened sugar maple health and made it more vulnerable to other stresses like climate extremes (St. Clair et al. 2008). Some studies even showed that thaw–freeze events can exacerbate nutrient deficiency on sugar maple (Comerford et al. 2013, Halman et al. 2013). Despite having identified thaw–freeze events, drought and forest tent caterpillar irruptions as possible causes for the 1986–1989

abrupt growth decrease, similar events did not cause such sudden growth decreases for sugar maple during the rest of the study period (Fig. 1A). For example, the three successive drought events in the mid-1970s did not have a drastic effect on sugar maple growth—even though they may have reduced its long-term vigor. Historical data are not precise enough to determine whether the 1986–1989 abrupt growth decrease was related to the intensity of one of the aforementioned events or the timing of their co-occurrence (e.g., thaw–freeze event, followed by a drought). Although it seems unlikely, it is also possible that an as yet unidentified stressor may be the cause of this 1986–1989 abrupt growth decrease. Nevertheless, whatever the exact causes of the 1986–1989 sugar maple abrupt growth decrease, it appears that it triggered irreversible damage (e.g., to the xylem tissue; Bréda et al. 2006), since sugar maple growth has still not recovered. More research is obviously needed to understand the causes of the 1986–1989 sugar maple abrupt growth decrease and the subsequent general decline. Again, it is possible that this decline be the result of an interaction of extreme events and acidic deposition.

The literature relating extreme events to forest ecosystems typically focuses on tree mortality. In some cases, the extreme climatic events, such as droughts or windstorms, may cause sudden death of a large number of trees (Hooper et al. 2001, Breshears et al. 2005). In other cases, increased rates of mortality are observed over a number of years after the extreme event, which often interacts with other biotic or abiotic stressors (McDowell et al. 2011, Amoroso et al. 2015, Camarero et al. 2015, Mamet et al. 2015). The case reported in this study is different as it is a reminder that the effects of extreme events may be more subtle and, even without leading to tree mortality, have important long-term consequences (Anderegg et al. 2015).

Subtle ecological effects that are induced by extreme events (hereafter, subtle effects) can be characterized by several particularities. First, subtle effects that modify physiological processes (Gutschick and BassiriRad 2003, McDowell et al. 2011) may have a long-term influence on species dynamics (Mamet et al. 2015) in forest ecosystems dominated by long-lived species as was observed for sugar maple growth, which

changed the competitive relationship between species. When individuals of different species (e.g., sugar maple vs. American beech) or different life stages within the same species (e.g., older vs. younger sugar maple) respond unequally, the long-term effects of the event will have a marked effect on the dynamics of the ecosystem (Mueller et al. 2005, Aakala et al. 2011).

Second, consideration of subtle effects adds perspective to the understanding of misadaptation, that is, current ecosystems may be poorly adapted to future climate conditions (Williams et al. 2007). Since sugar maple growth has not recovered from the 1986–1989 stress and its growth–climate relationship may have since shifted, this should be interpreted as a warning that this ecosystem may already be misadapted to the current climate. In other words, extreme climate events that generate subtle effects may accelerate ecosystem misadaptation to climate to a point where the resilience of the ecosystem is exceeded and from which the system drifts toward another basin of attraction (Walker et al. 2004).

Third, as the subtle effects that are caused by extreme events can last for long periods, there is a high probability that they will interact with other extreme events, stresses or disturbances to produce compound events (Denny et al. 2009) and decline spirals. This is perceptible in the larger sugar maple, for which the most vulnerable individuals in 2006–2007 are those that were the most severely affected during the 1986–1989 period. The consequences of such interactions between different stressors are difficult to predict. For example, for the sugar maple–American beech ecosystem, sugar maple decline co-occurs with the arrival of beech bark disease (Cale et al. 2017) which kills most mature beech trees. The co-occurrence of stressors that affects bigger stems of both species could lead to the development of dense, recalcitrant understory layers (Royo and Carson 2006), and the loss of forests with old-growth attributes that society frequently wishes to promote (Keeton 2006).

Our study highlights the challenge of predicting subtle effects, given that (1) causes are difficult to discriminate (e.g., thaw–freeze, drought, insects), (2) causes may have compound effects, (3) effects may vary among species and with ontogenic development, and (4) an initial stress may influence the response to further stress. In

the face of such uncertainty, monitoring ecosystem health is critical. This study illustrates that fine-scale monitoring is essential to complement broad-scale monitoring (Norman et al. 2016) as, for example, remote sensing or permanent sample plots that are re-measured every 5–10 yr could hardly detect changes in ecosystem dynamics as our targeted sampling had allowed us to do in this study. Moreover, as we expect that more and more ecosystems will be affected by subtle modifications in their dynamics, constant vigilance, both in the field and in data analyses, will be required to detect these changes.

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